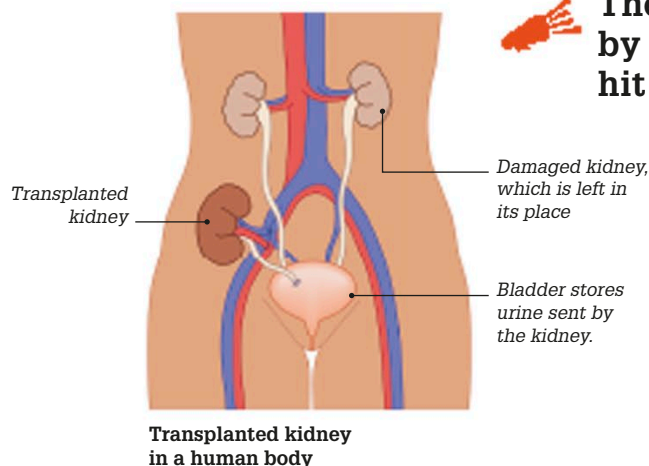


**1954****First organ transplant**

American surgeon Joseph Murray performed the first successful transplant of a human organ when he transplanted a kidney from one identical twin, Ronald Herrick, into the other, Richard. In this process, a transplanted kidney is implanted into the body and connected to blood vessels so that it can filter excess water and waste chemicals from the blood.

1955**1954****Polio vaccine trials**

The largest medical field trial ever began as 1.8 million schoolchildren were vaccinated in the US. They were given a vaccine that would protect them against the crippling disease of polio. The vaccine had been developed by American virologist Jonas Salk the previous year.

Randy Kerr, the first child to receive the vaccine during the field trial in 1954, stands beside Mary Kosloski, who had polio.



The first documented human to be struck by a meteorite was Ann Hodges, who was hit in her home in Alabama, in 1954.

1954 SOLAR CELL

A solar cell generates electric current when sunlight falls on it. Solar cells are made of photovoltaic materials. Inside these materials, electrons absorb the energy of the sunlight. This causes them to leave their atoms—and once freed, they flow through the material, creating an electric current. The first practical solar cell was developed at Bell Labs in the US in 1954. Today, solar cells have become a key source of renewable energy.

**First solar cell demonstration**

An executive from Bell Labs demonstrated the first practical solar cell in 1954. It generated enough electricity to power a 20.8-in- (53-cm-) high toy Ferris wheel and a small radio transmitter.

Vanguard I satellite

Launched in 1958, the US's second successful space satellite was also the first to be powered by solar cells, six of which were fitted to its body. The solar cells ran for almost seven years as the satellite orbited Earth.

Model of Vanguard I satellite, 1958

Solar cell provided power

Artificial diamond

American chemist Howard Tracy Hall created the first artificial diamond by heating carbon to incredibly high temperatures while under intense pressure—more than 100,000 times that of Earth's atmosphere.

